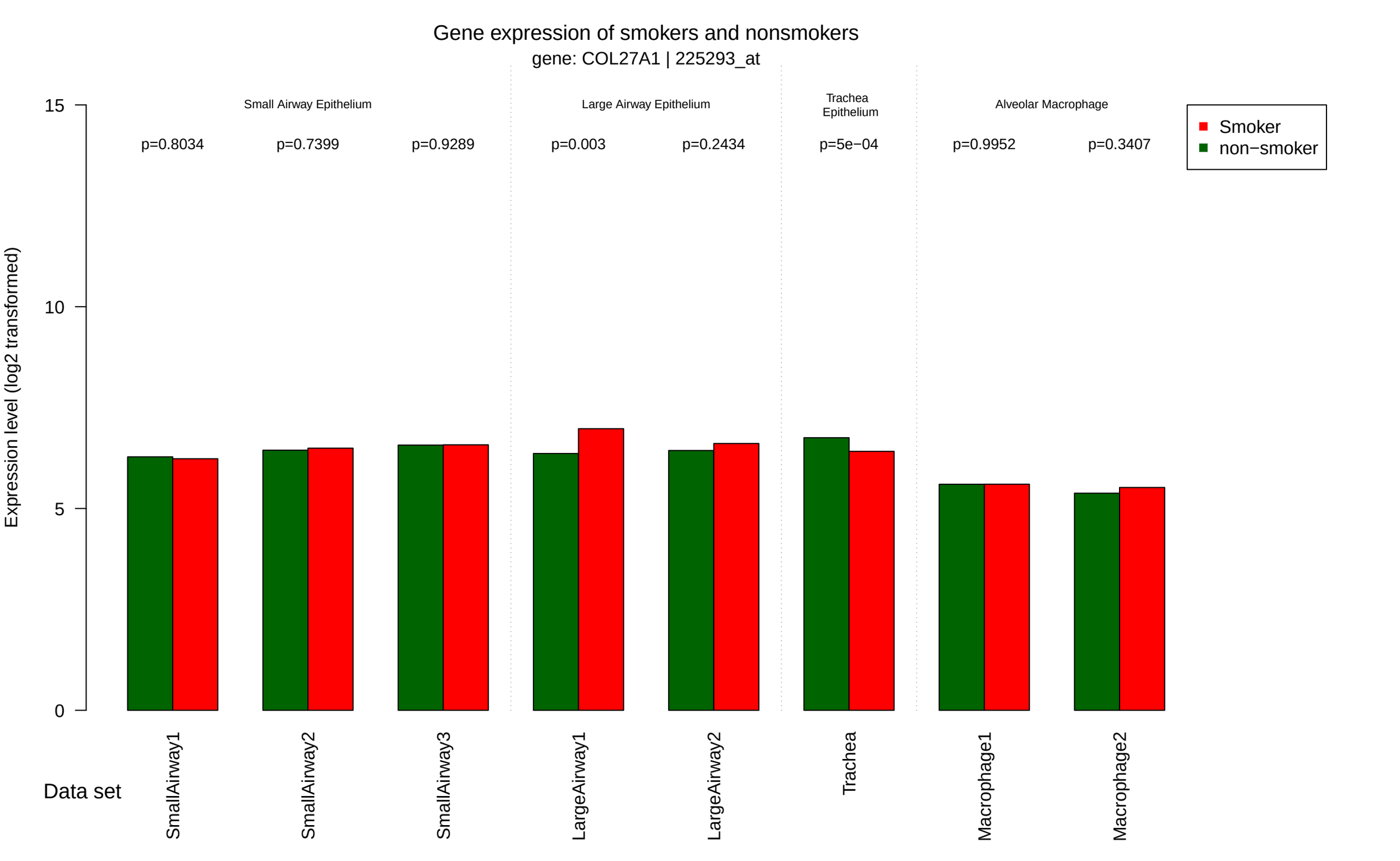
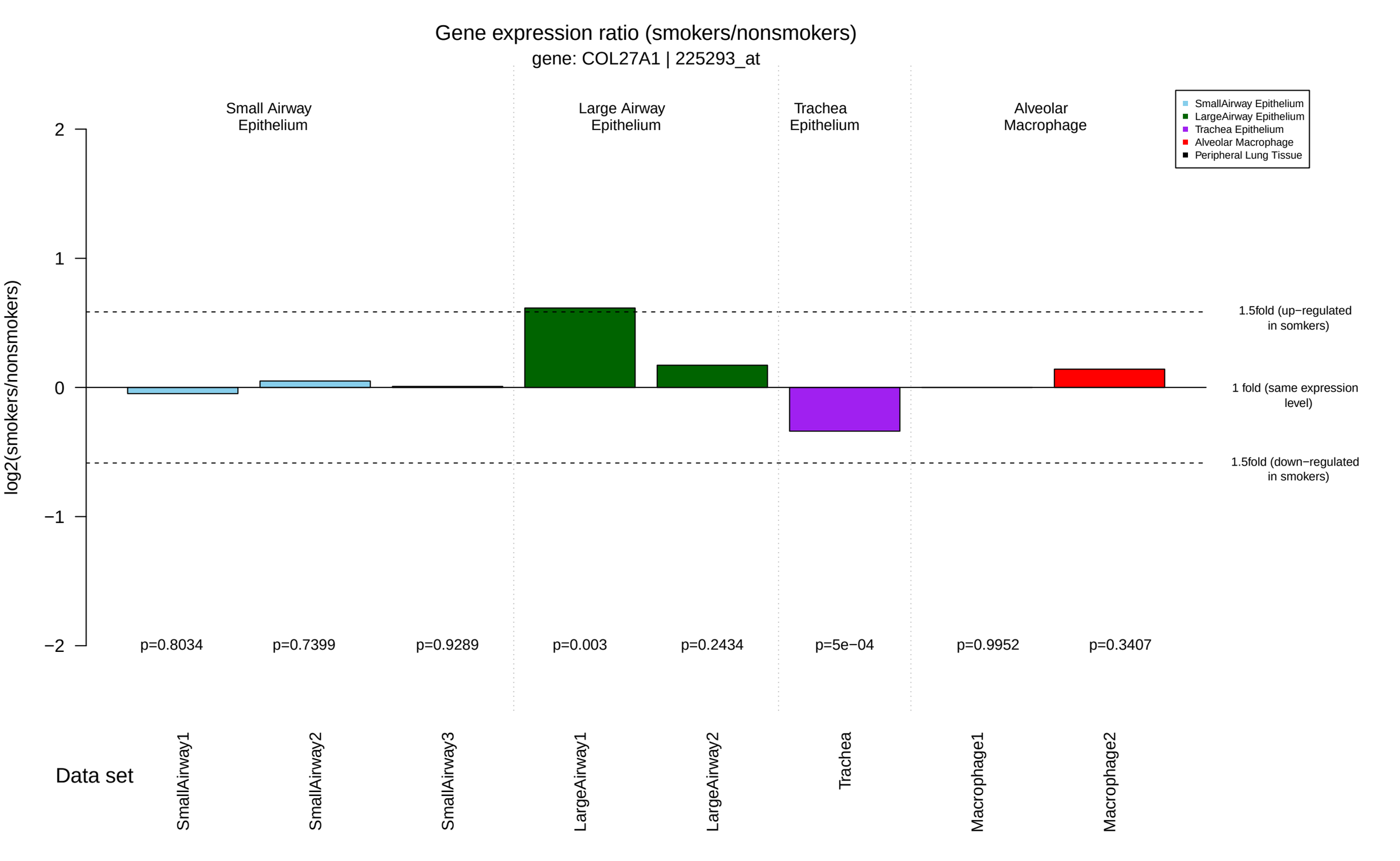
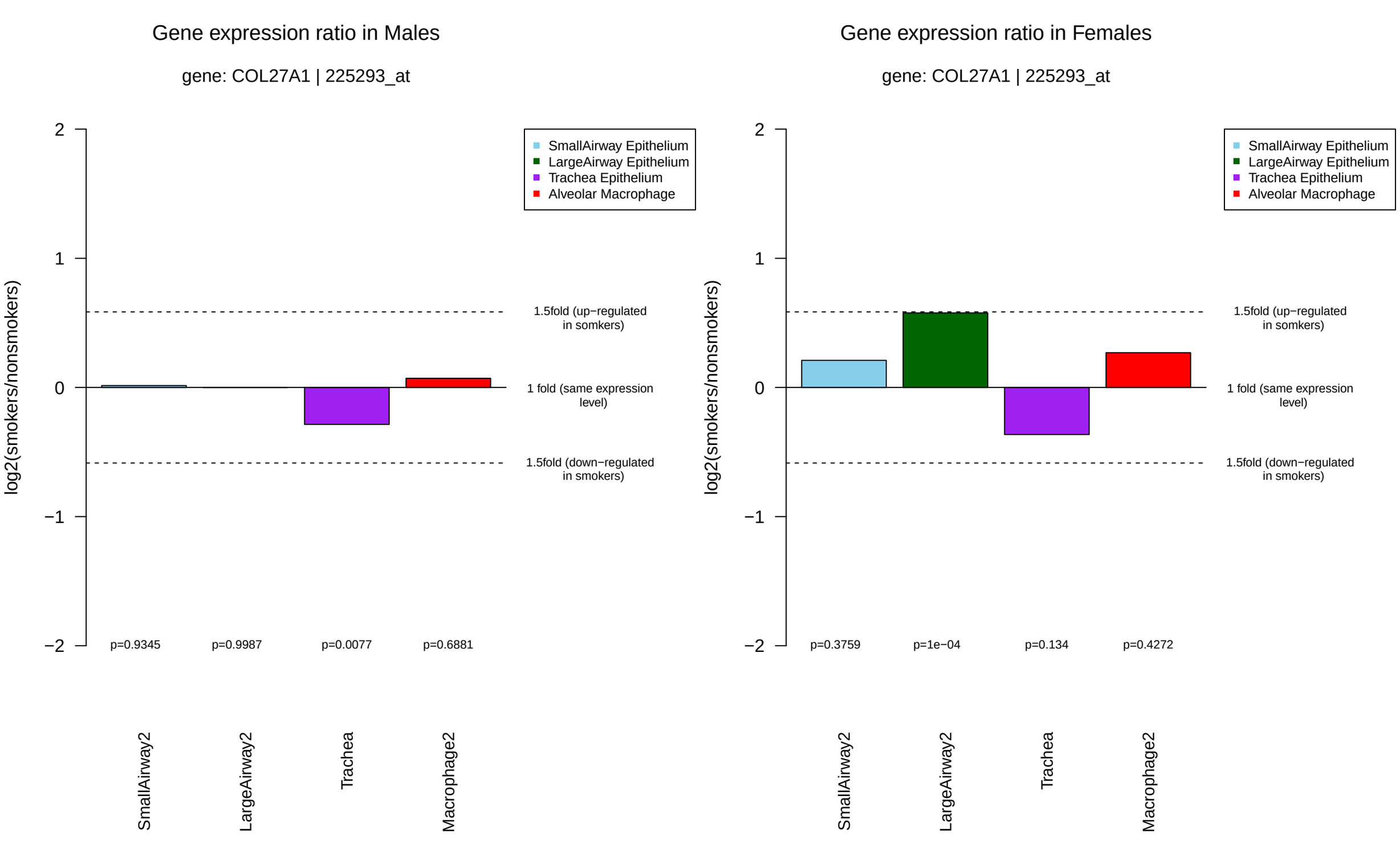
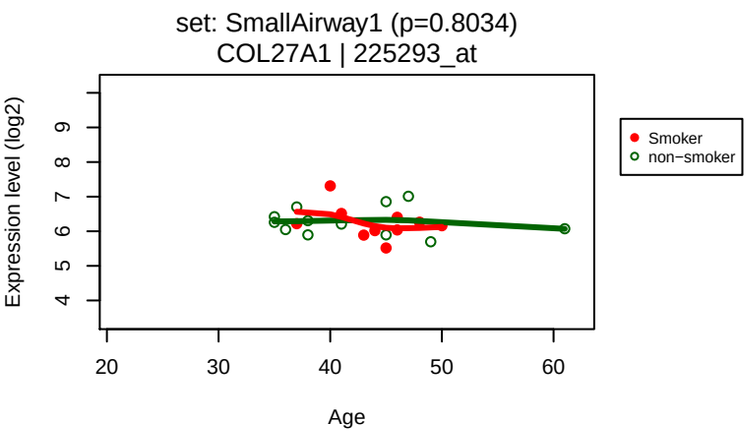
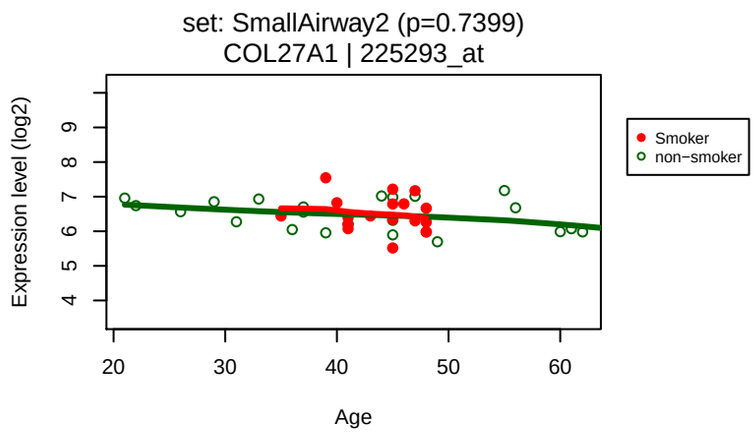
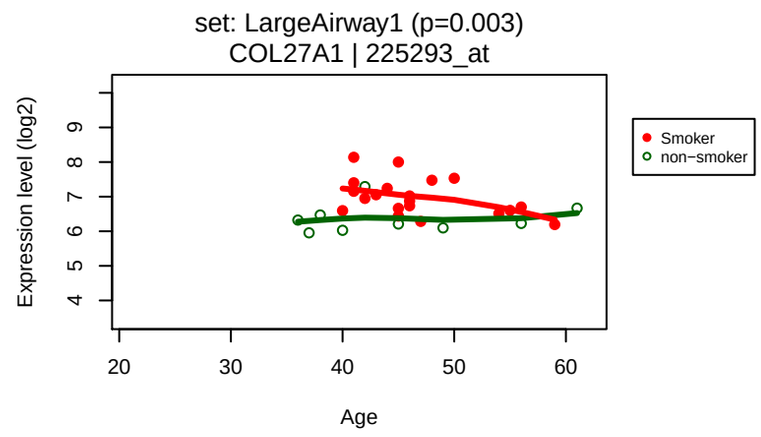

Gene expression of smokers and nonsmokers
gene: COL27A1 | 225293_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8034
p=0.7399
p=0.9289
p=0.003
p=0.2434
p=5e−04
p=0.9952
p=0.3407
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

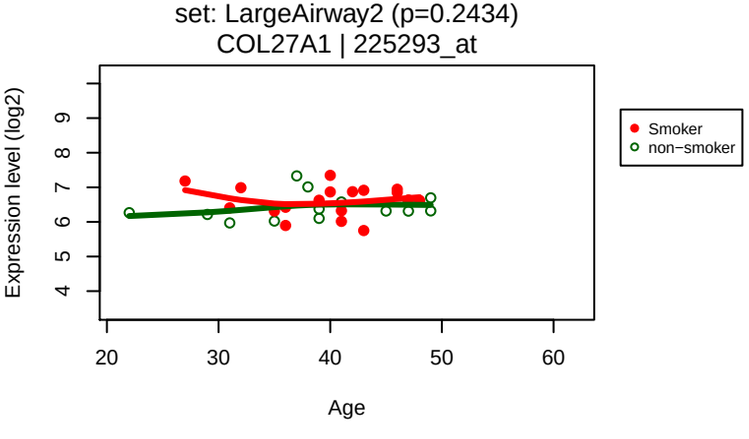

Gene expression ratio (smokers/nonsmokers)
gene: COL27A1 | 225293_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8034
p=0.7399
p=0.9289
p=0.003
p=0.2434
p=5e−04
p=0.9952
p=0.3407
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

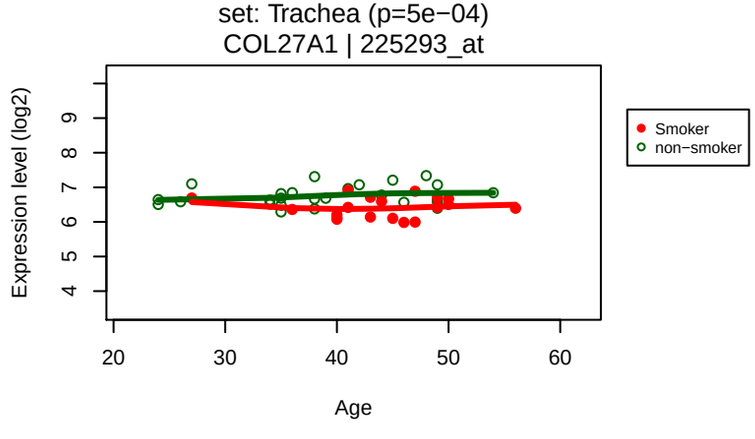

Gene expression ratio in Males
gene: COL27A1 | 225293_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9345
p=0.9987
p=0.0077
p=0.6881
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL27A1 | 225293_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3759
p=1e−04
p=0.134
p=0.4272
SmallAirway2
LargeAirway2
Trachea
Macrophage2

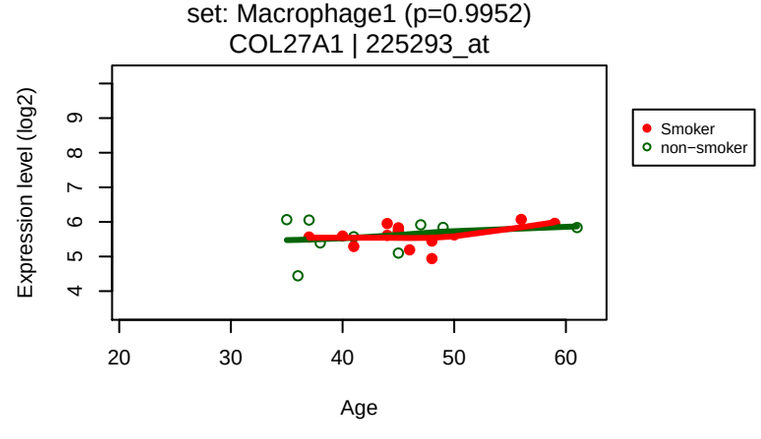

set: SmallAirway1 (p=0.8034)
COL27A1 | 225293_at
Expression level (log2)
Age
Smoker
non−smoker

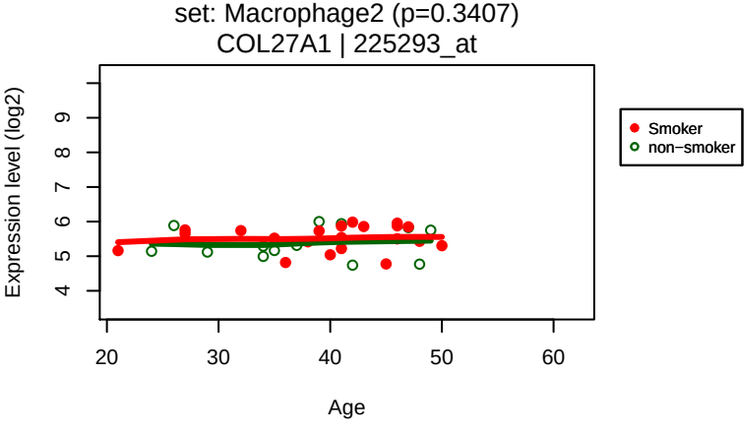

set: SmallAirway2 (p=0.7399)
COL27A1 | 225293_at
Expression level (log2)
Age
Smoker
non−smoker

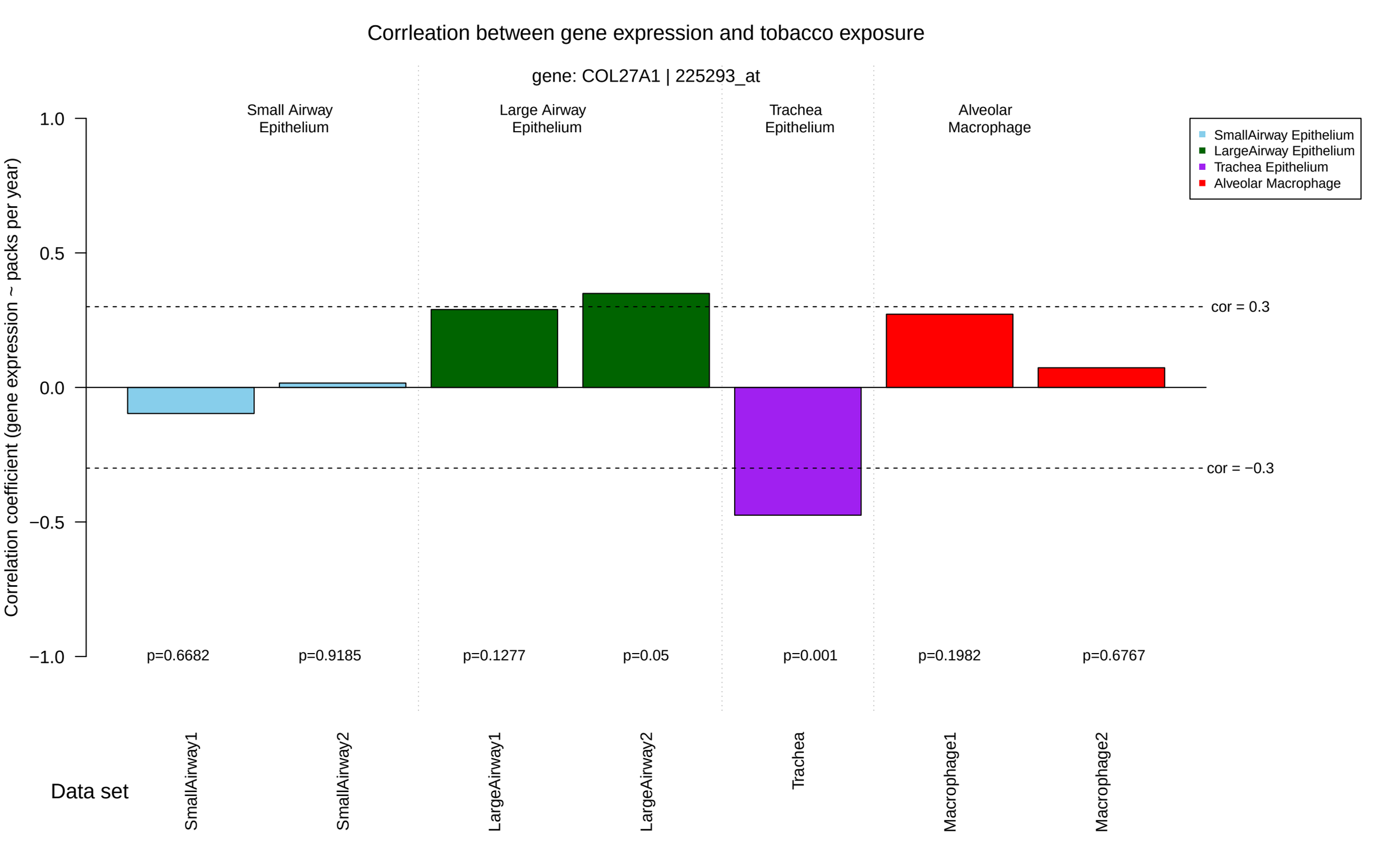

set: LargeAirway1 (p=0.003)
COL27A1 | 225293_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2434)
COL27A1 | 225293_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=5e−04)
COL27A1 | 225293_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9952)
COL27A1 | 225293_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3407)
COL27A1 | 225293_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL27A1 | 225293_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6682
p=0.9185
p=0.1277
p=0.05
p=0.001
p=0.1982
p=0.6767
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2